

PROYECTO 1



POKÉMON

PARTICIPANTES

01

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03

Calsina Gutierrez,
Treicy

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Macote Mattos,
James Brad

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Alvarez, Julian Edison



Antecedentes



smogon.com/dex/xy/pokemon/

02. Heading

GEN XY

SEARCH

POKEMON

MOVES

ABILITIES

ITEMS

In other gens: RB / GS / RS / DP / BW / XY / SM / SS / SV

Pokemon

FORMULA 1

TEMPORADA 2025

Plan Premium

Suscribirse

	Abomasnow	Grass	Ice	Snow Warning	Soundproof	NU	HP 90	Atk 92	Def 75	SpA 92	SpD 85	Spe 60
	Abomasnow-Mega	Grass	Ice	Snow Warning		RUBL	HP 90	Atk 132	Def 105	SpA 132	SpD 105	Spe 30
	Abra	Psychic		Synchronize	Inner Focus	LC	HP 25	Atk 20	Def 15	SpA 105	SpD 55	Spe 90
	Absol	Dark		Pressure	Super Luck	RU	HP 65	Atk 130	Def 60	SpA 75	SpD 60	Spe 75
	Absol-Mega	Dark		Magic Bounce		UU	HP 65	Atk 150	Def 60	SpA 115	SpD 60	Spe 115
	Accelgor	Bug		Hydration	Sticky Hold	RU	HP 80	Atk 70	Def 40	SpA 100	SpD 60	Spe 145



PROYECTO 2

IMPORTACIÓN



LECTURA DE DATOS



VOCABULARIO CONTROLADO

LIMPIEZA DE DATOS



TF-IDF



KMEANS



DATAFRAME

CSV FINAL



IMPORTACIÓN

ONE

TF-IDF

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

TWO

Pandas

```
import pandas as pd
```

THREE

KMeans

```
from sklearn.cluster import KMeans
```

© Let's Data Science

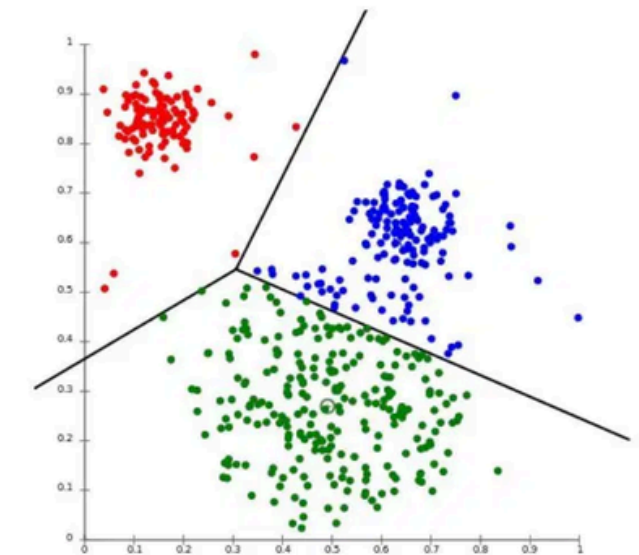


TF-IDF

Weighing Words in
Text Analytics



pandas





LECTURA DE DATOS

	Pokemon	url	texto	moves
0	Abomasnow	http://www.smogon.com/dex/xy/pokemon/abomasnow/	[{"group": "In other gens: DPBWXYSMAbomasnowIn-...	: NUUURUDoublesBattle Spot DoublesMovesAttrac...
1	Abra	http://www.smogon.com/dex/xy/pokemon/abra/	[{"group": "In other gens: RBGSRSDPBWXYSMAbraIn...	: LCMovesAlly SwitchPsychic Power—Accuracy—PP...
2	Absol	http://www.smogon.com/dex/xy/pokemon/absol/	[{"group": "In other gens: RSDPBWXYSMAbsolIn-ba...	: UURUMonotypeMovesAerial AceFlying Power60Ac...
3	Accelgor	http://www.smogon.com/dex/xy/pokemon/accelgor/	[{"group": "In other gens: BWXYSMAccelgorIn-bat...	: RUUUMovesAcidPoison Power40Accuracy100%PP30...
4	Aegislash	http://www.smogon.com/dex/xy/pokemon/aegislash/	[{"group": "In other gens: XYSMAegislashIn-batt...	: UberDoublesBattle Spot SinglesBattle Spot D...
...	...	...	...	...
751	Zoroark	http://www.smogon.com/dex/xy/pokemon/zoroark/	[{"group": "In other gens: BWXYSMZoroarkIn-batt...	: UUMovesAerial AceFlying Power60Accuracy—PP2...
752	Zorua	http://www.smogon.com/dex/xy/pokemon/zorua/	[{"group": "In other gens: BWXYSMZoruaIn-battle...	: LCMovesAerial AceFlying Power60Accuracy—PP2...
753	Zubat	http://www.smogon.com/dex/xy/pokemon/zubat/	[{"group": "In other gens: RBGSRSDPBWXYSMZubatI...	: LCMovesAcrobaticsFlying Power55Accuracy100%...
754	Zweilous	http://www.smogon.com/dex/xy/pokemon/zweilous/	[{"group": "In other gens: BWXYSMZweilousIn-bat...	: PUMovesAqua TailWater Power90Accuracy90%PP1...
755	Zygarde	http://www.smogon.com/dex/xy/pokemon/zygarde/	[{"group": "In other gens: XYSMZygardeIn-battle...	: OUDoublesMovesBindNormal Power15Accuracy85%...

756 rows × 4 columns

```
Pokemon = pd.read_csv("Project2/smogon.csv")
```





VOCABULARIO CONTROLADO

BUG



DARK



DRAGON



GROUND



STEEL



WATER





ELECTRIC



FAIRY



FIGHTING



FIRE



FLYING



GHOST



GRASS



Ice



POISON



PSYCHIC



ROCK



Limpieza de Datos

Data Original

	A	B
1		moves
2	0	: NUUURUDoublesBattle Spot DoublesMovesAttractNormal Power—Accuracy100%PP15Targets of the opposite gender are infatuated and have a 50% chance to do nothing.AvalanchePower60Accuracy100%PP10Base Power doubles if the user was damaged by the target previously in the same
3	1	: LCMovesAlly SwitchPsychic Power—Accuracy—PP15Switches the user's position with that of an ally. Priority +1.AttractNormal Power—Accuracy100%PP15Targets of the opposite gender are infatuated and have a 50% chance to do nothing.BarrierPsychic Power—Accuracy—PP20Boosts the user's
4	2	: UURUMonotypeMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.AssuranceDark Power60Accuracy100%PP10Base Power doubles if the target takes damage in the same turn before this move is used.AttractNormal Power—Accuracy100%PP15Targets of the
5	3	: RUUUMovesAcidPoison Power40Accuracy100%PP3010% chance to lower the target's Defense by one stage.Acid ArmorPoison Power—Accuracy—PP20Boosts the user's Defense by two stages.Acid SprayPoison Power40Accuracy100%PP20Lowers the target's Special Defense by two stages.Agi
6	4	: UberDoublesBattle Spot SinglesBattle Spot DoublesVGC14BHAAlmost Any AbilityMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.After YouNormal Power—Accuracy—PP15If the target has not yet moved, it will attack immediately after the user.AttractNormal
7	5	: UURUUberODoublesBattle Spot DoublesVGC14STABmonsMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.AgilityPsychic Power—Accuracy—PP30Boosts the user's Speed by two stages.Air CutterFlying Power60Accuracy95%PP25High critical hit rate.An
8	6	: UUNUOURUMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AttractNormal Power—Accuracy100%F
9	7	: LCMovesAcrobaticsFlying Power55Accuracy100%PP15Base Power doubles if the user has no held item.Aerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.AgilityPsychic Power—Accuracy—PP30Boosts the user's Speed by two stages.AstonishGhost Power30Accur
10	8	: OUAAlmost Any AbilityMovesAlly SwitchPsychic Power—Accuracy—PP15Switches the user's position with that of an ally. Priority +1.AttractNormal Power—Accuracy100%PP15Targets of the opposite gender are infatuated and have a 50% chance to do nothing.BarrierPsychic Power—Accuracy—PP
11	9	: RUUberOUUUMonotypeMovesAqua JetWater Power40Accuracy100%PP20Priority +1.Aqua RingWater Power—Accuracy—PP20User recovers 1/16th of max HP each turn.AttractNormal Power—Accuracy100%PP15Targets of the opposite gender are infatuated and have a 50% chance to do nothir
12	10	: PUOUNUBattle Spot DoublesMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.AgilityPsychic Power—Accuracy—PP30Boosts the user's Speed by two stages.Air CutterFlying Power60Accuracy95%PP25High critical hit rate.AstonishGhost Power30Accuracy1
13	11	: LCMovesAncient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AttractNormal Power—Accuracy100%PP15Targets of the opposite gender are infatuated and have a 50% chance to do nothing.Auro
14	12	: RUMovesAcrobaticsFlying Power55Accuracy100%PP15Base Power doubles if the user has no held item.Aerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.AgilityPsychic Power—Accuracy—PP30Boosts the user's Speed by two stages.AstonishGhost Power30Accur
15	13	: OUDoublesBattle Spot DoublesVGC16VGC14MovesAbsorbGrass Power20Accuracy100%PP25Leeches 50% of the damage dealt.After YouNormal Power—Accuracy—PP15If the target has not yet moved, it will attack immediately after the user.AstonishGhost Power30Accuracy100%PP1530% cha
16	14	: UUPUDoublesVGC14MovesAfter YouNormal Power—Accuracy—PP15If the target has not yet moved, it will attack immediately after the user.AgilityPsychic Power—Accuracy—PP30Boosts the user's Speed by two stages.AttractNormal Power—Accuracy100%PP15Targets of the opposite gender ai
17	15	: LCMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua JetWater Power40Accuracy100%PP20Priority +1.AttractNormal Power—Accuracy100%PP15Targets of the
18	16	: PUNUMovesAcidPoison Power40Accuracy100%PP3010% chance to lower the target's Defense by one stage.Acid SprayPoison Power40Accuracy100%PP20Lowers the target's Special Defense by two stages.Aqua TailWater Power90Accuracy90%PP10No additional effect.AttractNormal Power—A
19	17	: UUBattle Spot DoublesMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.AgilityPsychic Power—Accuracy—PP30Boosts the user's Speed by two stages.AttractNormal Power—Accuracy100%PP15Targets of the opposite gender are infatuated and have a 50%
20	18	: UberAGBHMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP
21	19	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
22	20	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
23	21	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
24	22	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
25	23	: UberAGMMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10I
26	24	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
27	25	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
28	26	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
29	27	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
30	28	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
31	29	: UberAGMMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10I
32	30	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
33	31	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
34	32	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
35	33	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
36	34	: UberMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10Bas
37	35	: UberAGMMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AvalanchePower60Accuracy100%PP10I
38	36	: LCMovesAcrobaticsFlying Power55Accuracy100%PP15Base Power doubles if the user has no held item.Aerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.AgilityPsychic Power—Accuracy—PP30Boosts the user's Speed by two stages.Ally SwitchPsychic Power—Ac
39	37	: NURUMovesAcrobaticsFlying Power55Accuracy100%PP15Base Power doubles if the user has no held item.Aerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.AgilityPsychic Power—Accuracy—PP30Boosts the user's Speed by two stages.Ally SwitchPsychic Power—
40	38	: PUMovesAgilityPsychic Power—Accuracy—PP30Boosts the user's Speed by two stages.AttractNormal Power—Accuracy100%PP15Targets of the opposite gender are infatuated and have a 50% chance to do nothing.Baton PassNormal Power—Accuracy—PP40The user switches, passing on all st
41	39	: PUMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua JetWater Power40Accuracy100%PP20Priority +1.Aqua TailWater Power90Accuracy90%PP10No additional
42	40	: RUUUDoublesBattle Spot DoublesMovesAfter YouNormal Power—Accuracy—PP15If the target has not yet moved, it will attack immediately after the user.AromatherapyGrass Power—Accuracy—PP5Cures status on the user's team.Aromatic MistFairy Power—Accuracy100%PP20The user raises it
43	41	: LCMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Ancient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.AttractNormal Power—Accuracy100%PP15Targets of the opposite gender are infatuated and have a 50% chance
44	42	: PURUNUBattle Spot SinglesMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.AgilityPsychic Power—Accuracy—PP30Boosts the user's Speed by two stages.Air CutterFlying Power60Accuracy95%PP25High critical hit rate.Ancient PowerRock Power60Accura
45	43	: NUPURUMovesAfter YouNormal Power—Accuracy—PP15If the target has not yet moved, it will attack immediately after the user.AmesiaPsychic Power—Accuracy—PP20Boosts the user's Special Defense by two stages.AttractNormal Power—Accuracy100%PP15Targets of the opposite gender a
46	44	: NURUMovesAncient PowerRock Power60Accuracy100%PP510% chance to boost all stats by one stage.Aqua TailWater Power90Accuracy90%PP10No additional effect.AttractNormal Power—Accuracy100%PP15Targets of the opposite gender are infatuated and have a 50% chance to do nothing.F
47	45	: PUMovesAfter YouNormal Power—Accuracy—PP15If the target has not yet moved, it will attack immediately after the user.AttractNormal Power—Accuracy100%PP15Targets of the opposite gender are infatuated and have a 50% chance to do nothing.AvalanchePower60Accuracy100%PP10Bas
48	46	: LCMovesAerial AceFlying Power60Accuracy—PP20Ignores accuracy and evasion modifiers.Aqua TailWater Power90Accuracy90%PP10No additional effect.AssuranceDark Power60Accuracy100%PP10Base Power doubles if the target takes damage in the same turn before this move is used.Attrac

hay mas de 48 filas



Pokémon



ERRORES en LOS DATOS

FireFire Po



CutterBug P



SlideRock



SOLUCIÓN

Fire Fire



Cutter Bug P



Slide Rock P





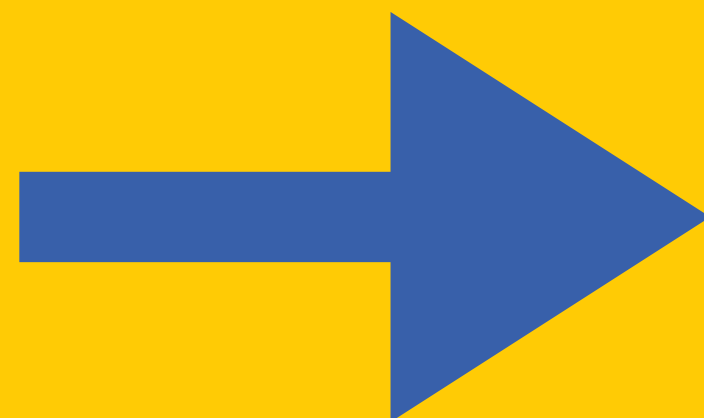
SOLUCIÓN

```
for TipoPokemon in VocabularioControlado:
    Pokemon["moves"] = Pokemon["moves"].str.replace(TipoPokemon, f" {TipoPokemon} ",
    regex=False)
MovimientoPokemon = Pokemon["moves"].str.replace(r"\s+", " ", regex=True).str.strip()
```

EJEMPLOS:

DATA =

index	tipo
0	fire
1	water
2	grass
3	electric
4	psychic
5	ice
6	dragon
7	dark
8	fairy
9	steel



DATA.STR.UPPER() =

index	tipo
0	FIRE
1	WATER
2	GRASS
3	ELECTRIC
4	PSYCHIC
5	ICE
6	DRAGON
7	DARK
8	FAIRY
9	STEEL



EJEMPLOS (CARACTERES ESPECIALES):

ReGex = FALSE

.RePLACE(“FIRE”, “FUEGO”, ReGex = FALSE)

Antes	Despues
"fire"	"agua"
"fireblast"	"aguablast"
"superfire"	"superagua"
"fire blast"	"agua blast"

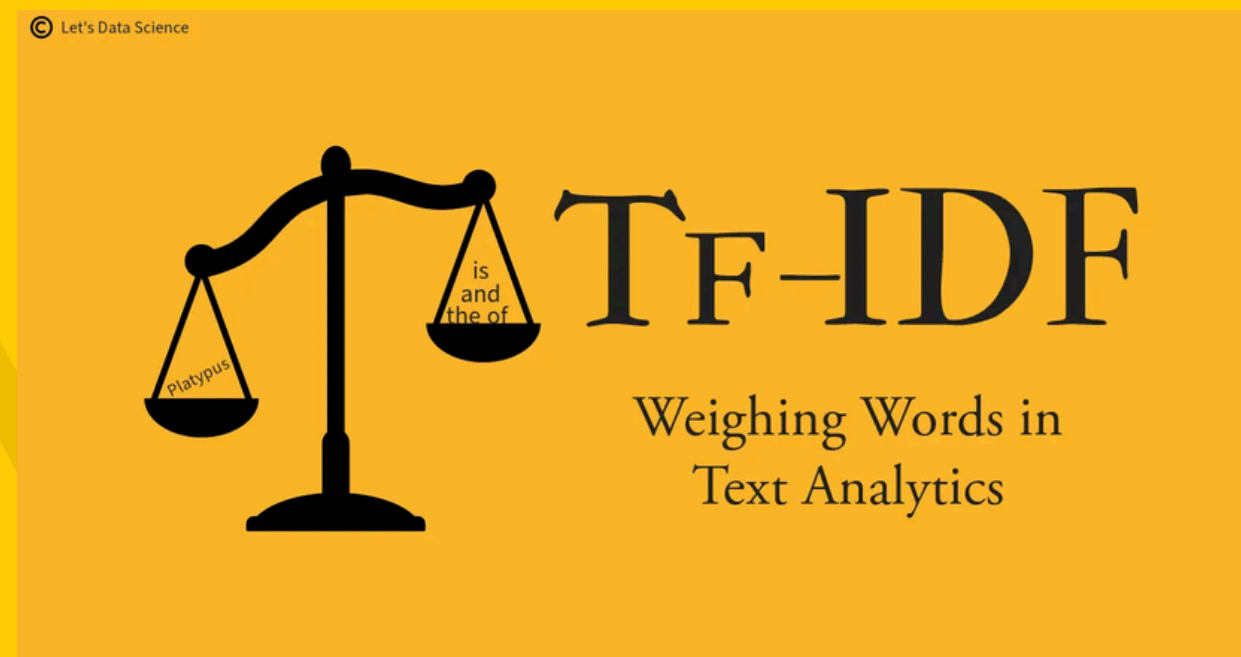
ReGex = TRUE

.RePLACE(“FIRE”, “FUEGO”, ReGex = TRUE)

Antes	Despues
"fire"	"agua"
"fireblast"	"fireblast"
"superfire"	"superfire"
"fire blast"	"fire blast"

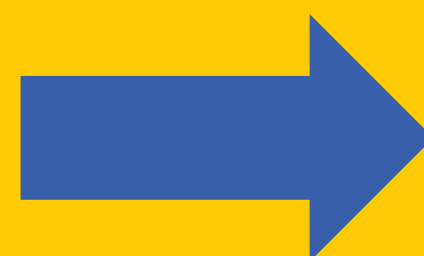


TF-IDF



UNIGRAMAS

**"PIKACHU es un POKEMON
DE TIPO FUEGO"**



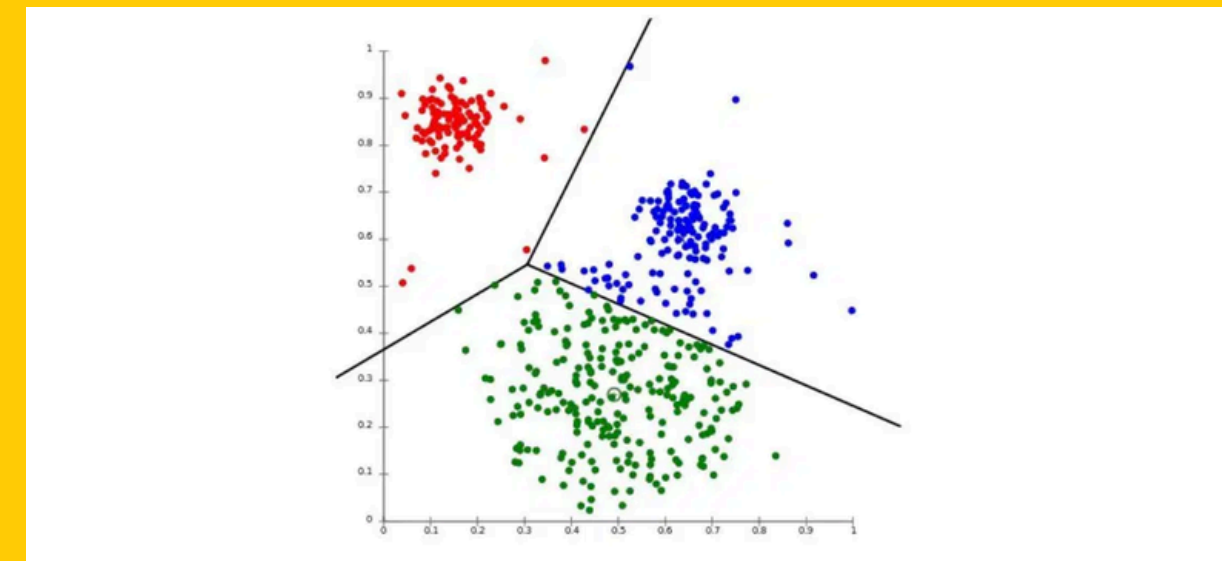
["PIKACHU", "es", "un", "POKEMON", "DE", "TIPO", "FUEGO"]



CLUSTERING

CLUSTERS:

17



N_INIT:

100

RANDOM_STATE:

42



DATAFRAME

	Pokemon	Cluster	Pokemon
0	Abomasnow	0	
1	Abra	11	
2	Absol	13	
3	Accelgor	10	
4	Aegislash	4	
...	...	...	
751	Zoroark	9	
752	Zorua	9	
753	Zubat	1	
754	Zweilous	14	
755	Zygarde	14	
756 rows × 2 columns			



Pokémon

GOTTA CATCH 'EM ALL!

